Thermodynamics

A fuel cell is a device that converts the chemical energy stored in a fuel into electricity. In a H_2/O_2 cell, the overall process is the formation of water. The process can be operated at low temperature (50-80 °C) or high temperature (120-180 °C). To simplify the calculation, we use the gas state for water.

$$H_2\,(g) + 1/2\,O_2\,(g) \longrightarrow H_2O\,(g)$$

divided into the two electrochemical reactions

$$\text{anode:} \quad H_2 \longrightarrow 2H^+ + 2e^-$$

$$\text{cathode:} \quad 1/2\,O_2 + 2H^+ + 2e^- \longrightarrow H_2O$$

The maximum work can be done by a fuel cell

$$W_{\max} = \Delta G = -nFE^0$$

where E^0 is the thermodynamic cell voltage

The thermodynamic efficiency of energy conversion in a fuel cell is defined as follows

$$\xi_{th} = \frac{\Delta G}{\Delta H} = 1 - \frac{T\Delta S}{\Delta H}$$

$$\text{or } \xi_{th} = \frac{E^0}{E_H^0} \text{ if } E_H^0 = -\Delta H / nF$$

Table 1. Thermodynamic data for compounds at 298 K.

	$\Delta_f H^o (kJmol^{-1})$	$\Delta_f G^o (kJmol^{-1})$	$S^o (Jmol^{-1}K^{-1})$
H_2 (g)	0	0	130.68
O_2 (g)	0	0	205.14
H_2O (g)	-241.82	-228.57	118.83

1. Calculate E^0, E_H^0, and thermodynamic efficiency (ξ_{th},%) of the H_2/O_2 cell at the operating temperature of 80 °C, based on the given thermodynamic data in Table 1. Assume ΔH and ΔS are independent of temperature. (30 pts)

2. Alternatively, chemical energy can be converted to heat through combustion to power a heat engine. For an automobile engine that burns diesel, the peak temperature of fuel-air mixture can reach around 900 °C. Assuming the cold sink temperature (T_{cold}) is close to the ambient temperature at 25 °C, calculate the maximum efficiency of a reversible heat engine based on Carnot cycle when it is operated between 900 °C and 25 °C. (20 pts)

3. Sketch plots of thermodynamic efficiency as a function of temperature ranging from 25 to 2000 °C for a reversible heat engine at constant T_{cold} of 25 °C and a H_2/O_2 fuel cell, respectively. Comment on how you would improve the thermodynamic efficiency with respect to temperature in each case. (40 pts)

4. In practice, efficiency of a heat engine is about 30-40%, while that of a H_2/O_2 cell is around 70%. Comment on the loss of energy in each case. (10 pts)